1 Label-free graph representation

Input graph
$G = (V, E, X)$

Node features X

Adjacency A

	v_1	v_2	⋯	v_n
v_1	0	1	⋯	0
v_2	1	0	⋯	1
⋯	⋮	⋮	⋱	⋮
v_n	0	1	⋯	0

Two augmented views

augmentation t_1: edge dropping + feature masking

augmentation t_2: independently sampled

GRACE shared GNN encoder g_ψ

projection head (contrastive training only)

Projected $Z^{(1)}$

Projected $Z^{(2)}$

Frozen normalized representation $\bar{H}$

After training: discard projection head; $\bar{H} = \mathrm{RowNorm}[g_\psi(X,A)]$

2 Label-agnostic granular-ball partition

Geometry-only construction in frozen embedding space $\bar{H}$

Initialization
$B_0 = V$

Recursive bi-partition

Further recursive splitting

Final granular-ball partition in $\bar{H}$
$\mathcal{B} = \{B_j\}_{j=1}^{K}$, $K \le B_{max} = 64$

selected ball B_j

All N nodes are partitioned using $\bar{H}$ only; labels do not affect construction. Splitting stops by the fixed budget or geometric criterion.

3 Leave-one-out ball-level label consensus

(a) Select a granular ball B_j

v's own vote removed

(b) Leave-one-out consensus

$p_{j,-v}$ (excluding v)

$$q_{j,-v} = (1-\beta)p_{j,-v} + \beta e_{m_{j,-v}}$$

For tied majorities, retain the empirical distribution $p_{j,-v}$.

(c) Construct a soft target t_v

noisy one-hot $e_{\tilde{y}_v}$ + local consensus $q_{j,-v}$ = soft target t_v

$$t_v = (1-\alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v}q_{j,-v}$$

$\alpha_{j,-v}$: vote concentration
$\beta = 0.5$ in the fixed configuration

4 X-only MLP training and inference

precomputed soft target t_v

MLP training with GB-LCSS targets

training node $v \in V_L$

node feature x_v

2-layer MLP f_θ

softmax

prediction $\hat{p}_v$

Soft cross-entropy
$\ell_v = -\sum_{c=1}^{C} t_{v,c} \log \hat{p}_{v,c}$

backpropagation: update θ

Transductive X-only inference (no graph access)

test-node feature x_u

trained MLP f_θ

softmax

class probabilities $\hat{p}_u$

Predicted class: 3
$\hat{y}_u = \arg\max_c \hat{p}_{u,c}$

Observed noisy labels (Stage 3 only): C1 C2 C3 C4 C5 — non-voting node — excluded self-vote — data flow — symmetric contrastive loss

X-only inference: no A, no $\bar{H}$, no neighbors